

Alexander Wang

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EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Computer Science & Bachelor of Business Administration

Aug. 2022 – May 2026

- Dual-Degree Cumulative GPA: 3.8/4.0
- Relevant Coursework: Data Structures & Algorithms (C++), Computer Architecture (C), Web Systems (JS, Python, Flask), Database Management Systems (SQL, NoSQL, JS), Intro to Artificial Intelligence (Python)
- Organizations: Sigma Eta Pi, Collaborative Lab for Advancing Work in Space, Michigan Fintech

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, C++, Python, Java, SQL, C#

Frameworks: React, Next.js, Spring Boot, Node.js, Flask, Swift, TensorFlow, scikit-learn, Pandas

Tools/Cloud: AWS (EC2, S3), GitHub Actions, MongoDB, MySQL, PostgreSQL, OracleDB, JUnit, Cypress, Git, Jira

Certifications: AWS Certified Cloud Practitioner (Aug. 2024), IBM Client Engineering Track (July 2024)

WORK EXPERIENCE

Software Engineering Intern

June 2025 – August 2025

Discover Financial Services

Riverwoods, IL

- Designed full-stack application replacement for Toolbox, Discover's legacy internal configuration management application for the core transaction switching platform; solely responsible for initiating the development process
- Built web console (React + TS/Tailwind, Vite, virtualized grid, 25-row pagination) and Cypress unit, component, & E2E testing; now reference design for team-wide UI migration, used by 10+ engineers across Payments and Network Authorizations org and expected to cut future build effort by 50%
- Wrote Spring Boot REST APIs (GET/POST/PATCH) with 100% JUnit coverage and corresponding script to auto-populate local MongoDB instance with 1200+ config entries from legacy Oracle DB, cutting deployment time to seconds and enabling real-time configuration edits within application web console

Software Engineering Lead

April 2024 – April 2025

Collage

Ann Arbor, MI

- Developed end-to-end social networking course selection platform in Next.js and Flask for college students as the lead developer and project manager, successfully scaling to over 750 users within 3 weeks of launch
- Shipped Google OAuth, social graph, schedule builder, activity feed; integrated Stripe metered billing in <3 days
- Migrated 1000+ lines of code from React SPA to Next.js 14 App Router, cutting Time-to-First-Byte by 38%
- Managed a cross-functional team of 10 members as a full time college student, coordinating development, design, and deployment to deliver the platform before Fall 2025 course registration

Software Engineering Extern

May 2024 – July 2024

IBM Accelerate

Los Angeles, CA

- Led team of 4 to develop a full-stack virtual assistant for insurance call center using IBM watsonx, designing 50+ intent-response flows to handle 9 high-volume inquiry categories; achieved 92% simulated resolution rate in testing
- Enhanced backend functionality by incorporating watsonx.ai NLP APIs for sentiment analysis, entity extraction, and summarization; applied prompt engineering techniques to reduce hallucination rate by 35% and improve intent classification accuracy by 20% in test conversations

PROJECT EXPERIENCE

Telepathy | Swift, XCode, Supabase (PostgreSQL + pgvector)

November 2025 - Present

- Built a native macOS overlay application that indexes users' Google Drive and Gmail into pgvector-powered vector embeddings, enabling semantic search and context-aware AI interactions across 20,000+ documents per user
- Implemented voice profile generation using Claude API to analyze user writing patterns, automatically injecting personalized style guides into AI conversations alongside connected documents and saved context
- Acquired 10 pilot customers and \$1,500 in funding through direct outreach and product iteration

Distributed Search Engine | Python, Flask, MapReduce

November 2024

- Built a Google-style search stack from scratch: engineered a five-stage MapReduce pipeline that parsed and indexed 150K Wikipedia pages (tf-idf + PageRank), then exposed shard results through a Flask REST API and multi-threaded aggregator UI, delivering top-10 results in < 100ms median latency